

Rainfall Trends in India Analysis: Overview and Dataset

Rainfall is critical in shaping India's economy, agriculture, and water resources. This analysis focuses on understanding the long-term trends, seasonal patterns, and anomalies in rainfall across the country. By examining historical rainfall data spanning over a century, we aim to:

- Examine historical rainfall trends
- Detects year with extreme or deficient rainfall
- Explore the relationship between seasonal and annual rainfall
- Forecast future rainfall using a time series model

The dataset used for this analysis is comprehensive, providing monthly rainfall measurements, seasonal aggregations (such as monsoon rainfall), and annual totals. These metrics are crucial for studying rainfall variability, understanding extreme events, and predicting future trends. The dataset's rich temporal coverage allows for a robust evaluation of rainfall dynamics over time, making it a valuable resource for addressing the challenges posed by climate change and rainfall variability in India.

Why this project?

Understanding rainfall patterns is vital for India due to its significant impact on agriculture, water resource management, and disaster preparedness. Over 50% of the country's population depends on agriculture, which relies on seasonal rainfall, particularly the monsoon. Erratic rainfall patterns, including droughts and floods, can disrupt livelihoods, food security, and the economy.

This project aims to:

- **Address Climate Concerns:** With climate change influencing weather patterns, identifying long-term trends and anomalies in rainfall can help assess its potential effects.
- **Improve Decision-Making:** Insights from this analysis can guide policymakers in developing sustainable water resource management strategies and disaster mitigation strategies.
- **Support Agricultural Planning:** Understanding seasonal rainfall variability aids farmers in making informed decisions about crop selection and irrigation.
- **Advance Research:** By forecasting future rainfall trends, this project contributes to the growing body of knowledge on climate science and its implications.

The project is driven by a desire to better understand rainfall dynamics and provide data-driven insights that can benefit multiple sectors, ensuring better resilience and adaptability to changing weather patterns.

About the project:

This project focuses exclusively on the analysis of historical rainfall data to uncover patterns, trends, and anomalies without integrating a front-end or user interface. By leveraging a comprehensive dataset containing over a century of monthly, seasonal, and annual rainfall measurements, the study aims to provide valuable insights into India's rainfall variability and its long-term behavior.

The project is designed as a standalone data analysis endeavor, with an emphasis on extracting meaningful insights through statistical techniques and time series modeling. While it does not involve web or application-based integration, the results aim to contribute to the understanding of rainfall dynamics, supporting research and decision-making in areas such as climate adaptation, agriculture, and water resource management.

For this analysis, we rely on a dataset containing monthly, seasonal, and annual rainfall measurements spanning over a century. I found an ideal dataset for this task, which includes monthly rainfall values, seasonal aggregations (e.g., monsoon rainfall), and annual totals, all of which are essential for understanding variability, detecting anomalies, and predicting future trends. Link for the dataset:

<https://statso.io/rainfall-trends-in-india-case-study/>

Rainfall Trends in India Analysis with Python

Now, we will begin by loading and exploring the dataset containing rainfall data for India from 1901 to 2015. Each record provides monthly, seasonal, and annual rainfall measurements. This data is critical for understanding long-term trends, seasonal variations, and extreme weather events:

```
import pandas as pd

rainfall_data = pd.read_csv(r"C:\Users\sakhi\Downloads\rainfall_area-wt_India_1901-2015.csv")

print(rainfall_data.head())
```

	REGION	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	\
0	INDIA	1901	34.7	37.7	18.0	39.3	50.8	113.4	242.2	272.9	124.4	
1	INDIA	1902	7.4	4.3	19.0	43.5	48.3	108.8	284.0	199.7	201.5	
2	INDIA	1903	17.0	8.3	31.3	17.1	59.5	118.3	297.0	270.4	199.1	
3	INDIA	1904	14.4	9.6	31.8	33.1	72.4	164.8	261.0	206.4	129.6	
4	INDIA	1905	25.3	20.9	42.7	33.7	55.7	93.3	252.8	200.8	178.4	

		OCT	NOV	DEC	ANNUAL	Jan-Feb	Mar-May	Jun-Sep	Oct-Dec
0		52.7	38.0	8.3	1032.3	72.4	108.1	752.8	99.0
1		61.5	27.9	24.4	1030.2	11.7	110.8	794.0	113.8
2		117.9	36.9	17.7	1190.5	25.3	107.9	884.8	172.5
3		69.0	11.2	16.3	1019.8	24.0	137.4	761.8	96.6
4		51.4	9.7	10.5	975.3	46.2	132.2	725.4	71.6

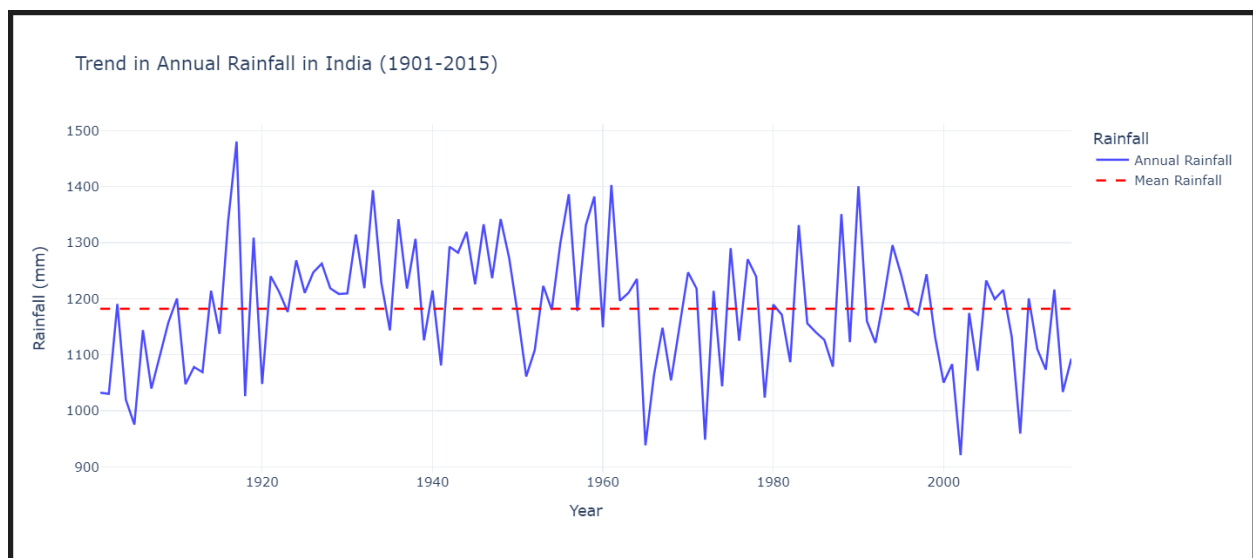
In the above data, the region indicates the geographical area of INDIA. The year represents the year in which the data is recorded. This also represents monthly rainfall values in millimeters. ANNUAL gives us the total annual rainfall. There is also seasonal rainfall for different intervals of different months. The monthly columns (JAN to DEC) contain rainfall data in millimeters for each month of the year.

Analyzing Annual Rainfall Trends Over Time

Rainfall is one of the most critical components of Earth's water cycle and plays a vital role in shaping ecosystems, supporting agriculture, and maintaining human livelihoods. However, rainfall patterns have never been constant—they vary naturally due to climatic conditions and are now increasingly influenced by human-induced climate change.

Analyzing annual rainfall trends over time is a key step in understanding how these patterns are changing. By examining long-term rainfall data, researchers and policymakers can identify trends such as increases or decreases in total rainfall, shifts in seasonal contributions, or the emergence of extreme weather events. These insights are crucial for effective water resource management, disaster preparedness, and planning for agriculture and infrastructure development.

In regions like India, where agriculture heavily depends on monsoons, rainfall variability can have significant social and economic impacts. Long-term datasets, such as the annual rainfall data from 1901 to 2015, offer an opportunity to study historical trends, detect anomalies, and assess the potential effects of climate change on precipitation patterns.



In the above picture, the blue line in the graph represents yearly recorded rainfalls in millimeters for India. The fluctuations in the blue line indicate variability in the annual rainfall from year to year. Some years received higher rainfall (peaks), while others experienced lower rainfall (troughs).

The Red dashed line in the graph represents average rainfall over the entire period (1901–2015). This is a reference line that helps compare how each year's rainfall deviates from the long-term average.

The X-axis in the graph shows the timeline from 1901 to 2015, allowing us to observe trends over more than a century.

The Y-axis (rainfall in mm) in the graph indicates the amount of annual rainfall recorded, ranging roughly from 900 mm to 1500 mm.

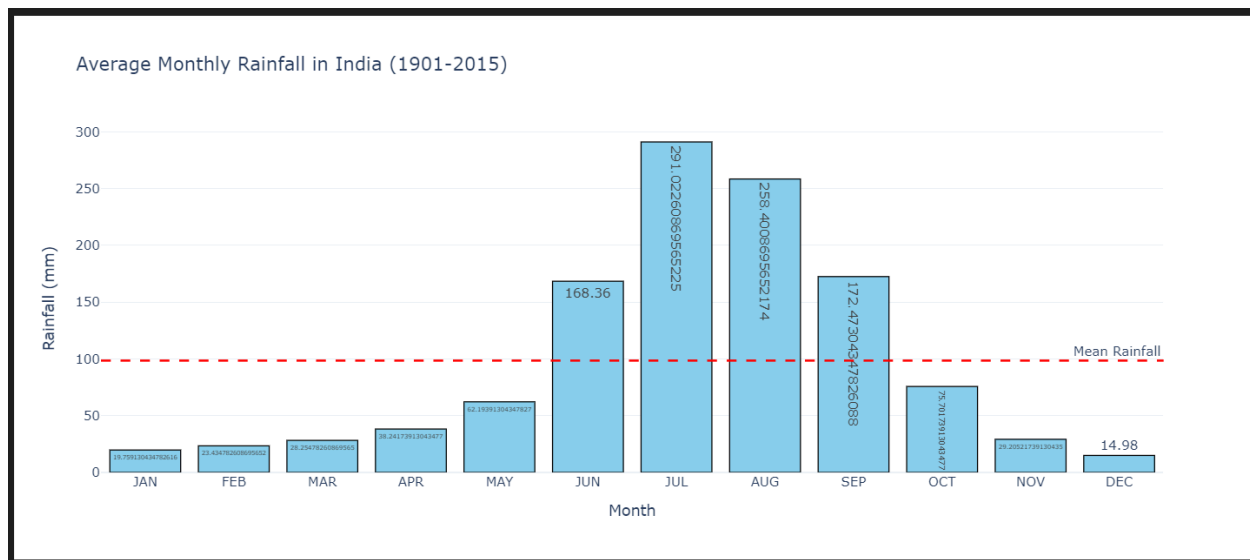
Observations:

The blue line shows significant ups and downs in annual rainfall, reflecting natural variability in weather patterns. Some extreme years stand out with either very high or very low rainfall.

Peaks (e.g., around the 1910s and mid-1960s) indicate years with unusually high rainfall. Troughs (e.g., the early 1900s and late 1970s) indicate years with deficient rainfall.

The red dashed line (mean rainfall) shows that the average rainfall remained relatively stable across the 115 years. This suggests that while annual rainfall fluctuates, the overall average does not drastically change.

Though the mean rainfall is stable, there might be a slight decrease in the amplitude of the fluctuations after the mid-20th century (1960s onwards). This could suggest a reduction in extreme events, but further analysis is needed to confirm this.



In the above picture of bar graphs, each bar represents the average rainfall (in millimeters) recorded in a particular month from 1901 to 2015. The height of the bar indicates the amount of rainfall, with taller bars showing higher averages for that month.

The red dashed line in the graph represents the overall monthly mean rainfall across all months. It acts as a reference to compare individual months' rainfall against the overall average.

The X-axis(month) displays the months of the year, from January (JAN) to December (DEC), making it easy to track seasonal changes.

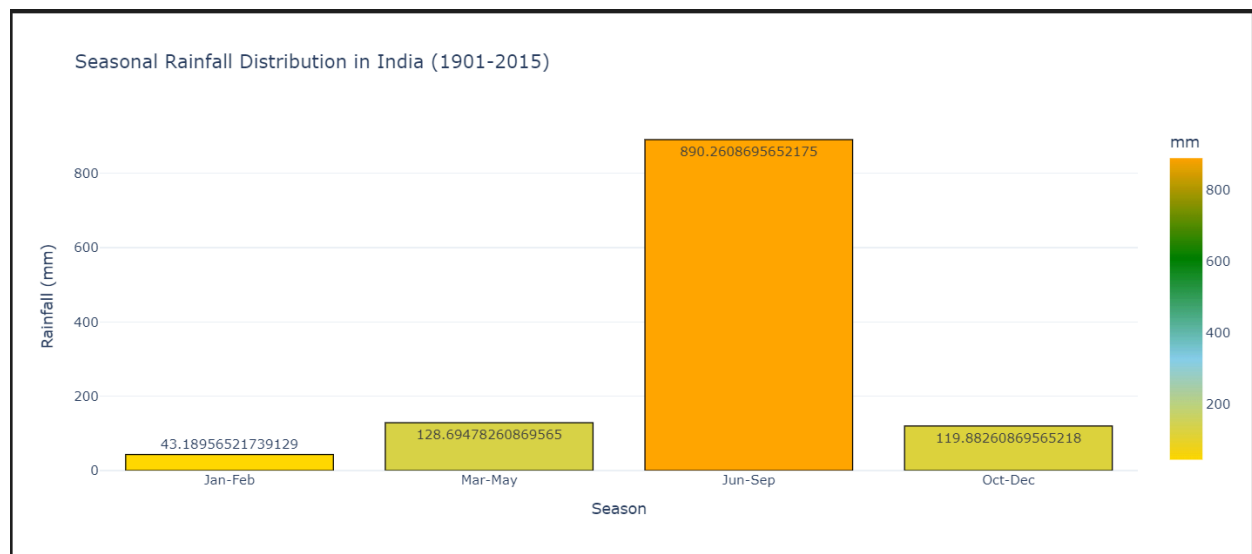
Y-axis(Rainfall in mm) indicates the amount of rainfall in millimeters (mm), ranging from 0 mm to over 300 mm.

Observations:

Rainfall is not evenly distributed across the months. June, July, August, and September (monsoon months) receive the highest rainfall, with July having the peak average rainfall of 291.02 mm. In contrast, the months of November, December, January, and February have much lower averages, with December recording the least rainfall at just 14.98 mm.

The months of June to September collectively account for the majority of India's rainfall. This aligns with the importance of the monsoon season for the country. The sharp increase in rainfall from May to June and the gradual decline from September to October reflect the onset and withdrawal of the monsoon. Dry periods are seen from October to May, with significantly lower rainfall. Wet periods dominate from June to September, driven by monsoon activity.

Months like July and August are well above the red dashed line, indicating they contribute significantly more rainfall than the overall monthly average. Months like January, February, November, and December fall well below the red line, highlighting their low contribution to the annual rainfall.



In the above picture of the bar graph, each bar represents the average rainfall (in millimeters) for a specific season. The height and color intensity of the bars indicate the total amount of rainfall, with deeper colors (e.g., orange) signifying higher rainfall.

The X-axis(Season) represents the seasons are grouped into four intervals:

- Jan-Feb: Winter
- Mar-May: Pre-monsoon (summer)
- Jun-Sep: Monsoon
- Oct-Dec: Post-monsoon (retreating monsoon)

The Y-axis(Rainfall in mm) indicates the rainfall amount in millimeters, ranging from 0 to over 800 mm, making it easy to compare the rainfall between seasons.

The number on the bars displays the exact average rainfall values for each season, providing precise measurements for interpretation.

The color legend on the right side explains the color intensity, where brighter shades (yellow) represent lower rainfall and darker shades (orange) represent higher rainfall.

Observations:

The monsoon season (Jun-Sep) receives the highest rainfall, with an average of 890.26 mm, making it the most significant contributor to India's annual rainfall. This period is crucial for agriculture, water reservoirs, and replenishment of natural water bodies.

The winter season (Jan-Feb) records the lowest average rainfall of 43.19 mm, highlighting its dry nature. This aligns with the general expectation of low precipitation during this time, except for some regions affected by western disturbances.

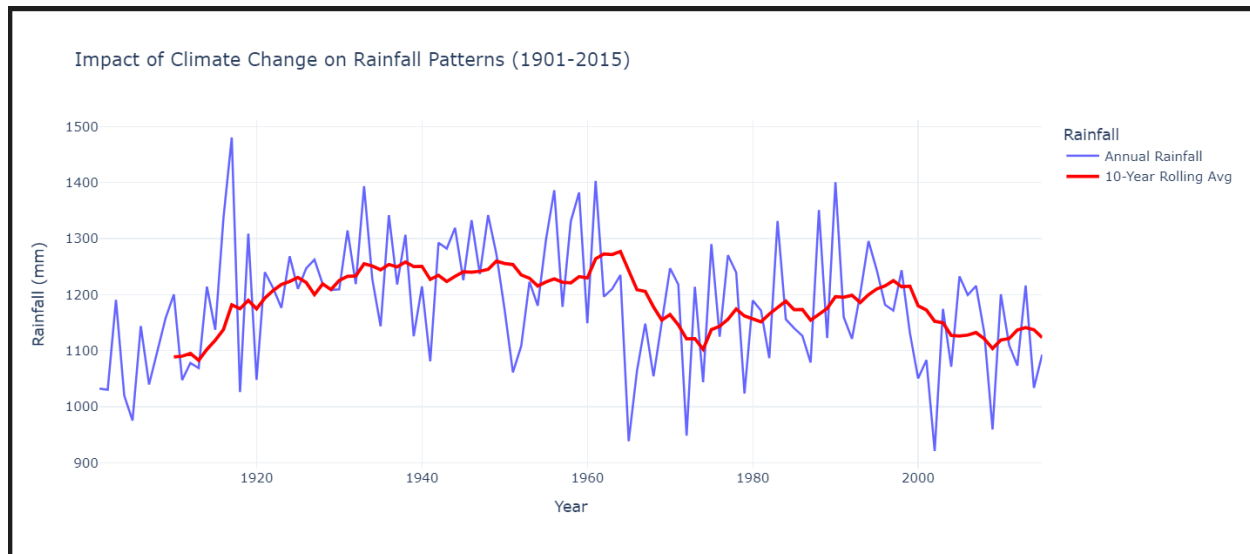
The pre-monsoon season (Mar-May) records an average rainfall of 128.69 mm. This rainfall is primarily caused by localized thunderstorms and cyclonic activities in some parts of India.

The post-monsoon season (Oct-Dec) shows an average rainfall of 119.88 mm, which is slightly lower than the pre-monsoon period. Rainfall during this period is influenced by the retreating monsoon and cyclonic systems, especially over southern and eastern India.

The monsoon season accounts for the overwhelming majority of rainfall, while the remaining three seasons contribute comparatively little. The difference between the monsoon (890.26 mm) and the second-highest season, pre-monsoon (128.69 mm), highlights India's heavy reliance on monsoon rains.

Assessing the Impact of Climate Change on the Rainfall Trends in India

Now, we will calculate a 10-year rolling average of annual rainfall to identify long-term trends and smooth out short-term variations.



This graph shows the annual rainfall trends in India (blue line) and a 10-year rolling average (red line) to identify long-term patterns. While annual rainfall exhibits significant variability, the 10-year rolling average indicates a slight downward trend post-1960, which suggests a possible impact of climate change on rainfall distribution. Periods of higher averages in the early 20th century contrast with more consistent but lower averages in recent decades.

Observations:

Rainfall shows significant fluctuations from year to year, indicating the influence of natural phenomena such as monsoon variability, El Niño, La Niña, and other climate factors. Some years experience extreme rainfall (peaks above 1400 mm), while others see reduced rainfall (dips below 1000 mm).

Annual rainfall starts below 1100 mm, with a gradual increase observed. The red line shows a slight upward trend during this period, indicating an increase in average rainfall.

The rainfall stabilizes, with the 10-year rolling average hovering around 1200 mm. Periodic peaks and troughs indicate wet and dry years but no significant long-term change during this time.

A noticeable decline in the red line begins after the 1960s, indicating a reduction in average rainfall. The decrease aligns with concerns about climate change and human-induced impacts on weather patterns.

The rainfall appears to stabilize again around 1100–1200 mm, though the red line remains slightly lower than mid-century levels. The rolling average suggests no immediate recovery to the higher rainfall levels of earlier decades.

Now, using statistical thresholds (1.5 standard deviations below or above the mean), let's identify years with extreme or deficient rainfall.

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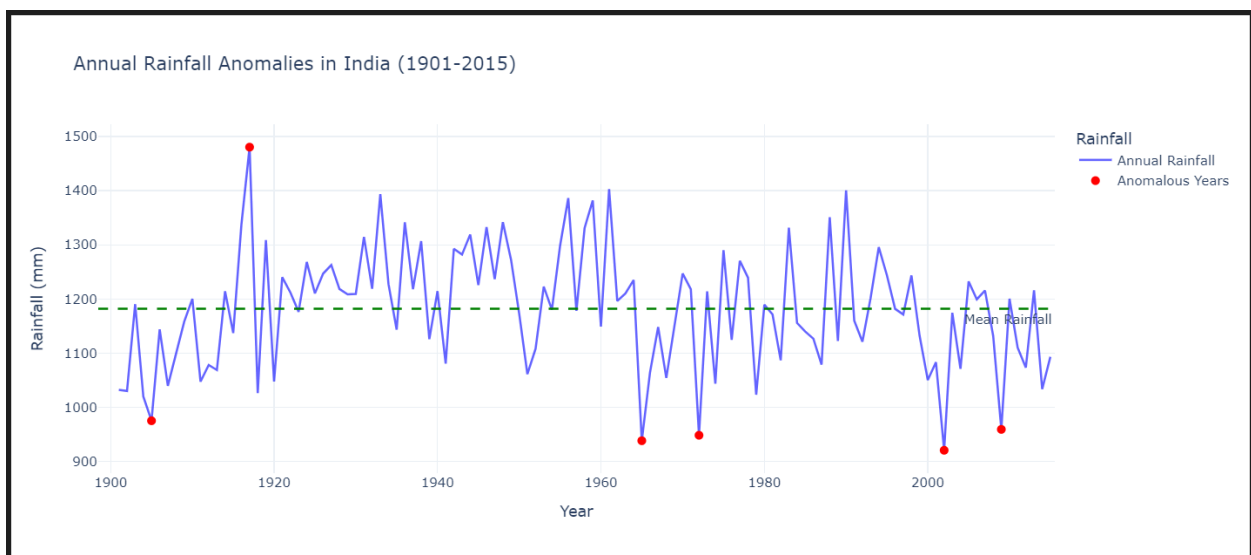
... ( YEAR ANNUAL
0 1905 975.3
1 1965 938.4
2 1972 948.5
3 2002 920.8
4 2009 959.3,
YEAR ANNUAL
0 1917 1480.3
1 1933 1393.5
2 1956 1386.2
3 1959 1382.1
4 1961 1403.0
5 1988 1351.0
6 1990 1400.6,
Correlation
Jan-Feb 0.228913
Mar-May 0.313057
Jun-Sep 0.930027
Oct-Dec 0.531648)

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The analysis identifies five significant drought years (e.g., 2002 and 2009) and seven extreme rainfall years (e.g., 1917 and 1990) based on deviations from the mean annual rainfall. Seasonal rainfall correlations with annual totals reveal that the monsoon season (June-September) has the strongest correlation (0.93), which indicates it predominantly drives annual rainfall patterns. In contrast, other seasons like January-February (0.23) and March-May (0.31) have weaker correlations, which emphasizes the critical role of the monsoon in India's overall rainfall dynamics.

Detecting Anomalies in the Rainfall Trends in India

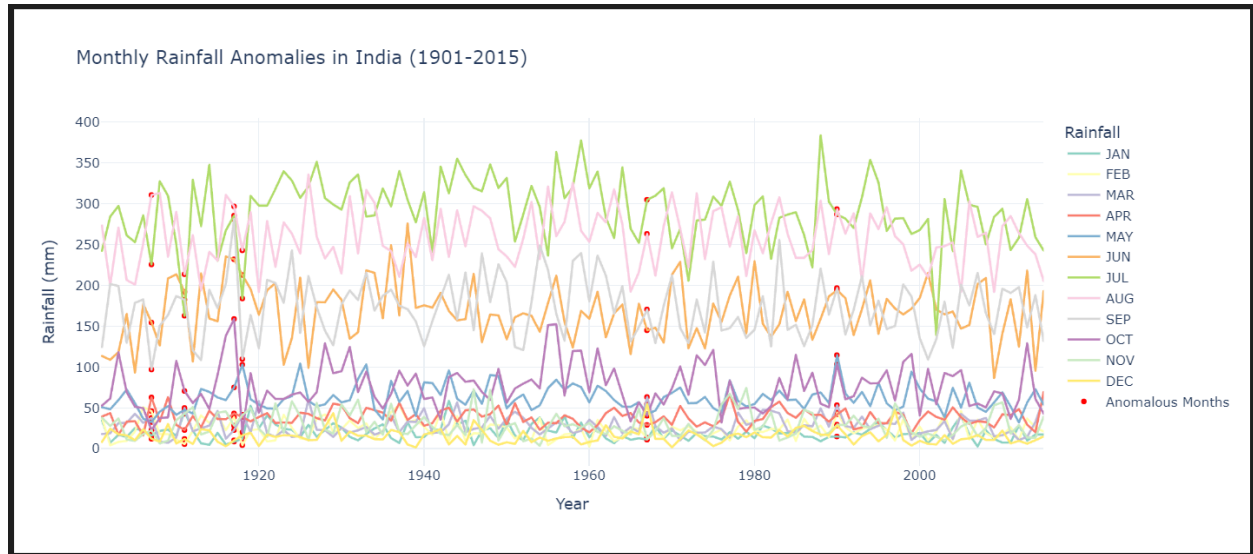
Now, using an Isolation Forest algorithm, we will identify anomalies in both annual and monthly rainfall. This will highlight specific years or months with unusual rainfall patterns. Let's start with identifying anomalies in annual rainfall:



This graph highlights years with significant rainfall anomalies, where annual rainfall deviated substantially from the mean. Drought years (e.g., 1905, 1965, 2002) and extreme rainfall years (e.g.,

1917, 1961) are marked as red points, which showcase outliers in rainfall patterns. While most years cluster around the mean (green dashed line), the anomalies emphasize the variability in India's rainfall, driven by factors like monsoonal fluctuations and climate events. This underscores the need for monitoring and preparedness for extreme weather events.

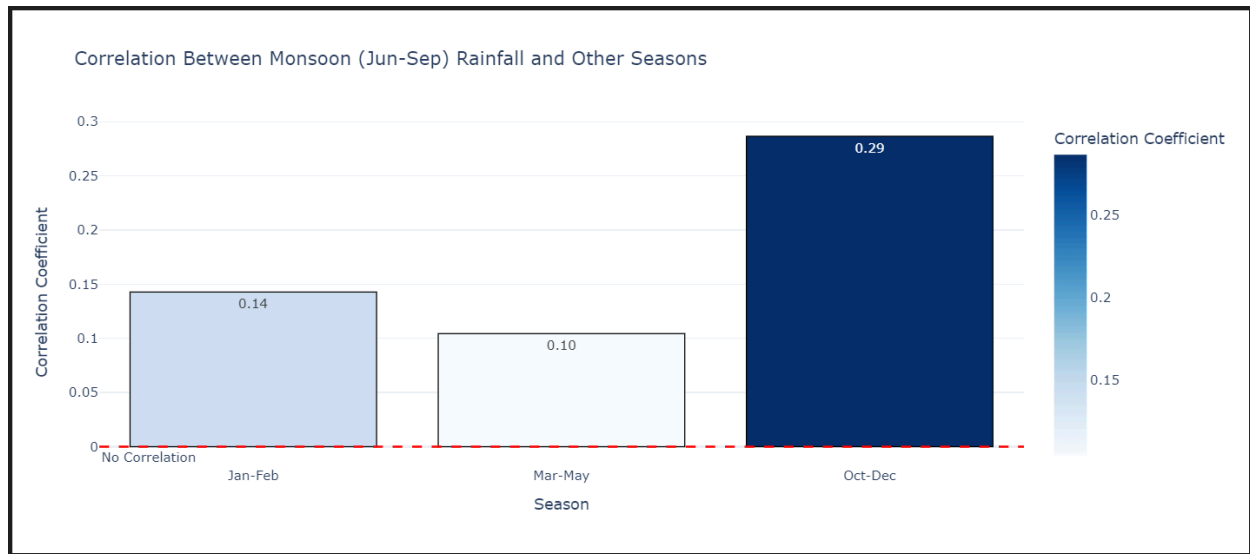
Now, let's identify anomalies in monthly rainfall:



The variability is most pronounced during the monsoon months (June to September), which reflects the critical role of these months in India's rainfall dynamics. Anomalies in non-monsoon months, while less frequent, highlight periods of unusual weather patterns, potentially linked to climate variability or regional disturbances. This graph underscores the uneven distribution and high dependence on monsoonal rainfall for India's water resources.

Correlating Seasonal Rainfall with Annual Totals

Now, we will calculate the correlation coefficients between seasonal rainfall and annual rainfall totals to understand how much each season contributes to the overall yearly rainfall:



This graph shows the correlation between monsoon rainfall and rainfall during other seasons. The October-December season has the highest correlation (0.29), which suggests a moderate relationship, possibly due to the post-monsoon retreat rains. The January-February (0.14) and March-May (0.10) seasons exhibit weaker correlations, which indicate minimal dependence on monsoon rainfall. This highlights the dominance of monsoonal patterns as an independent driver of India's annual rainfall, with limited spillover effects on other seasons.

Observations:

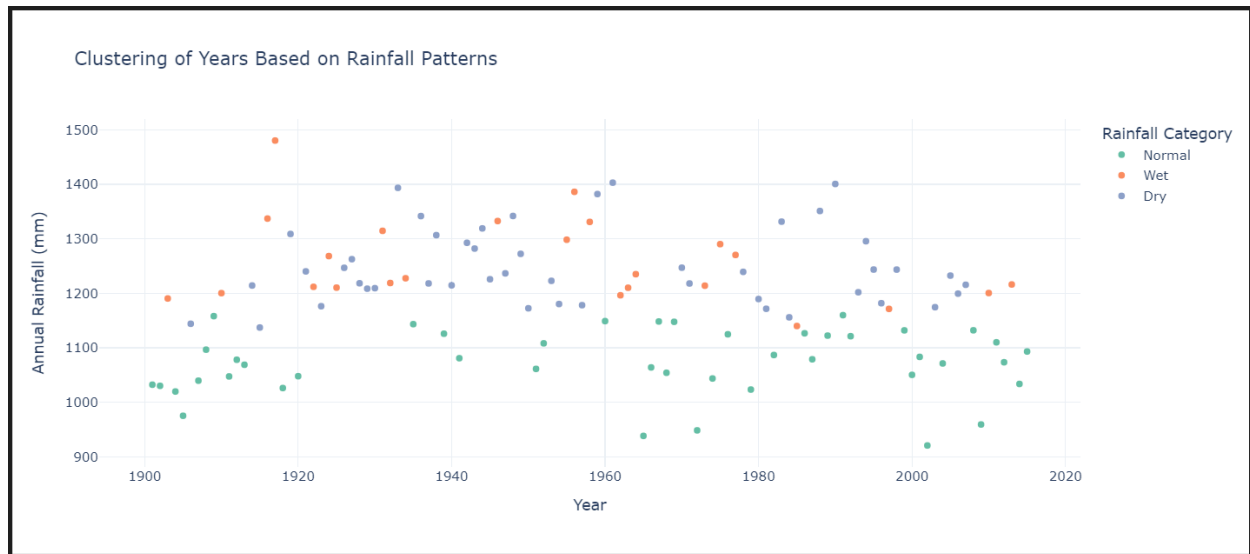
In Jan-Feb (Winter) correlation coefficient was 0.14. Which indicates a weak positive relationship between monsoon and winter rainfall. This means that rainfall during the monsoon season has a small influence on rainfall in the winter months, but the connection is not strong.

In Mar-May (Pre-Monsoon) correlation coefficient was 0.10. Suggests a very weak positive relationship between monsoon and pre-monsoon rainfall. This indicates almost no dependence on rainfall in these two seasons.

In Oct-Dec (Post-Monsoon) correlation coefficient was 0.29. Shows the strongest positive correlation among the three seasons. This suggests that monsoon rainfall has a slightly more noticeable impact on rainfall during the post-monsoon season, likely due to the extended influence of monsoon systems transitioning into the post-monsoon period.

Grouping Years Based on Rainfall Patterns

Now, by applying k-means clustering, we will group years into three categories: Dry, Normal, and Wet, based on rainfall patterns:



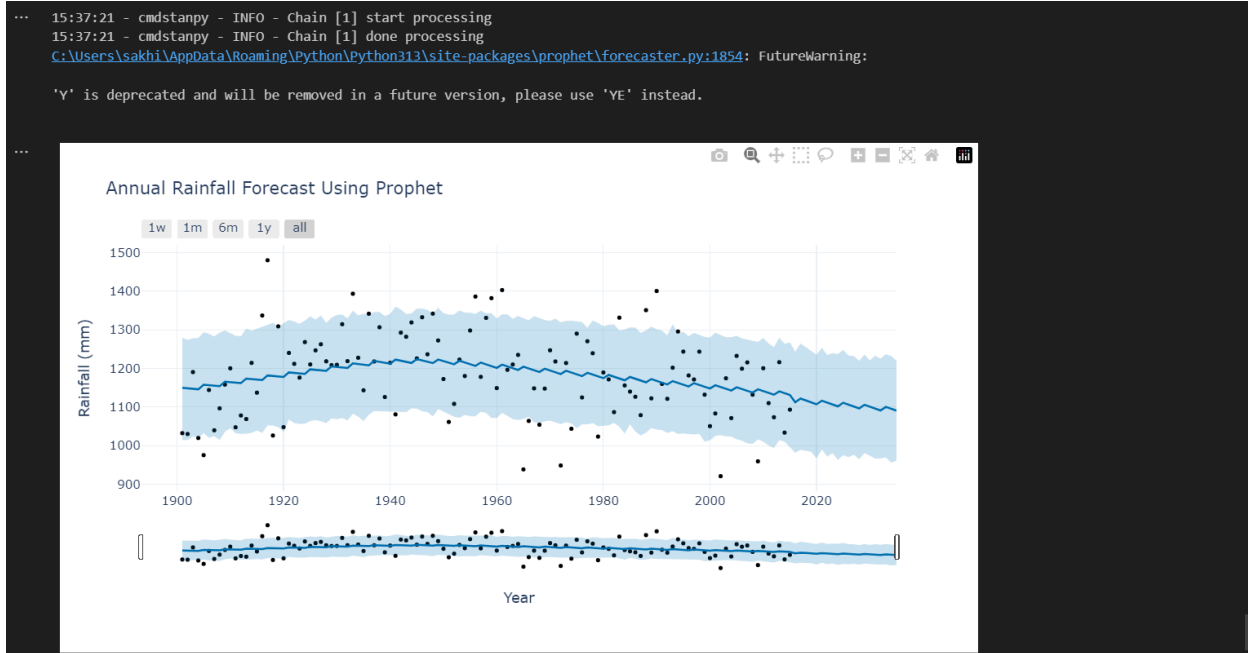
The clusters reveal that most years fall into the Normal category, while Wet years (above-normal rainfall) are sporadically distributed throughout the timeline, with a concentration in the early and mid-20th century. Dry years (below-normal rainfall) are more frequent in the latter half of the timeline, which indicates a potential shift in rainfall patterns over time. This clustering emphasizes the variability and potential long-term changes in India's rainfall dynamics.

Observations:

Most years fall within the normal range, suggesting a steady pattern over time. Wet years (orange dots) and dry years (blue dots) are scattered throughout the timeline, showing variability in extreme conditions. From 1900 to 1930, rainfall is mostly normal (green) or slightly dry (blue), with a few wet years. From 1950 to 1970, there are clusters of wet years, indicating an unusually rainy period. After 2000, a significant number of years are classified as dry, suggesting a possible drying trend. Rainfall fluctuates significantly, with dry years dipping below 1100 mm and wet years exceeding 1400 mm.

Forecasting Future Rainfall

We will use the Prophet Library to forecast annual rainfall for the next 20 years:



The blue line represents the model's forecast trend, while the shaded area indicates the confidence interval. The trend reveals a slight decline in annual rainfall over time, with notable year-to-year variability (black dots representing actual data points). The model captures the variability well but highlights that future rainfall may continue to slightly decrease, which emphasizes the need for adaptive strategies to manage potential water resource challenges.

Observations:

From around 1900 to the mid-20th century, the trend shows a slight increase in annual rainfall. After 1960, there was a gradual decline in rainfall, as indicated by the downward slope of the blue line. The confidence intervals widen slightly in earlier years, reflecting uncertainty in older historical data. After 2020, the forecast becomes less certain, shown by a broader blue-shaded area. Rainfall appears to decrease steadily after 2000, both in observed data and in the forecast. The black dots show significant variability year-to-year, with some extreme years (very high or very low rainfall).

Conclusion:

The analysis of India's rainfall trends and patterns from 1901 to 2015 reveals significant variability in annual and seasonal rainfall, with the monsoon season (June-September) being the dominant contributor, accounting for the majority of the country's annual rainfall. Anomalous years of extreme drought and wetness highlight the unpredictability of rainfall, while clustering shows a discernible shift towards more

frequent dry years in recent decades. This trend raises concerns about the increasing vulnerability of agricultural and water-dependent sectors to prolonged dry spells.

Correlations between monsoon and non-monsoon seasons indicate the limited dependency of non-monsoon rainfall on monsoon patterns, suggesting that rainfall in these seasons is driven by independent climatic factors such as Western Disturbances or localized thunderstorms. The relatively stronger correlation between monsoon and post-monsoon (Oct–Dec) rainfall points to the residual effects of monsoon systems, which could be critical for regions relying on post-monsoon precipitation for water supply and cropping.

A time-series forecast using the Prophet model suggests a slight declining trend in annual rainfall, signaling long-term challenges for water availability and agricultural productivity. This declining trend, coupled with the increasing frequency of anomalous years, underscores the urgent need for strategic water resource planning, improved irrigation infrastructure, and adoption of climate-resilient agricultural practices.

Additionally, the variability in rainfall patterns calls for enhanced meteorological monitoring, predictive modeling, and early warning systems to mitigate the impacts of extreme weather events. Policymakers must prioritize investment in adaptive measures, such as rainwater harvesting, drought-resistant crop varieties, and regional-level water-sharing frameworks, to build resilience against the impacts of changing climate patterns.

This comprehensive understanding of rainfall trends is crucial for guiding sustainable development policies, ensuring food security, and safeguarding livelihoods, particularly in rural areas heavily dependent on monsoon rains. The findings emphasize the critical need for integrating climate adaptation strategies into national and regional planning to address the challenges posed by evolving rainfall patterns.